

Mesenchymal Stem Cell Exosome-Mediated Delivery of Paclitaxel for Pancreatic Cancer Therapy

Biomolecules

1. Introduction

Pancreatic cancer is one of the most lethal solid tumors, accounting for approximately 8% of cancer-related deaths in the United States, despite representing only about 3% of all cancer cases [1]. The five-year survival rate is approximately 13%, the lowest among major cancers. This dismal prognosis is primarily due to late detection, as early-stage pancreatic cancer typically lacks specific symptoms and reliable diagnostic markers [2]. Consequently, most cases are identified at advanced or metastatic stages, with only about 15% of patients eligible for surgical intervention, so most rely on chemotherapy [3]. The standard first-line treatments of pancreatic cancer include gemcitabine, alone or with other drugs like Abraxane (albumin-bound paclitaxel) or FOLFIRINOX (a regimen of folinic acid, 5-fluorouracil, irinotecan, and oxaliplatin) [4]. However, the therapeutic efficacy of conventional chemotherapeutics is often limited by poor tumor selectivity [5,6], severe toxicity [7,8], and drug resistance [9,10,11], highlighting the necessity for advanced delivery strategies that increase drug accumulation at tumor sites while reducing off-target effects [12].

Nanocarrier-based systems, including lipid-based [13,14,15], polymeric [16,17,18] and inorganic nanoparticles [19,20], micelles [21,22], and dendrimers [23,24], have emerged as effective platforms for controlled release, enhanced bioavailability, and extended circulation times [5,16,25]. Despite these advantages, significant challenges remain, including achieving precise spatiotemporal targeting, mitigating the potential toxicity of carrier materials, and overcoming biological barriers such as the immune clearance mechanisms and tumor heterogeneity [26]. Moreover, clinical translation is further hindered by issues such as reproducible large-scale manufacture, biocompatibility, and the complexity of the tumor microenvironment [27]. Therefore, innovative strategies are urgently needed to enhance tumor specificity, biocompatibility, and overall therapeutic outcomes.

In recent years, exosomes have attracted growing interest as biogenic nanovesicles for targeted drug delivery, offering distinct advantages over conventional synthetic nanocarriers such as liposomes and polymeric nanoparticles [28,29,30,31]. Their endogenous origin confers high biocompatibility, low immunogenicity, and minimal cytotoxicity, while their structural stability enables efficient cargo protection and delivery [32,33]. Moreover, exosomes possess intrinsic targeting capabilities, allowing them to traverse biological barriers, evade immune surveillance, and facilitate selective uptake by recipient cells. Among various sources, MSC-derived exosomes have emerged as particularly promising due to their inherent tumor-homing capabilities [34,35]. Unlike whole-cell therapies, MSC-derived exosomes mitigate the risks associated with uncontrolled cell proliferation or differentiation, representing a safe, cell-free alternative for next-generation cancer treatments [36].

In this study, we explore the feasibility and therapeutic potential of MSC-derived exosomes as an efficient nanocarrier for paclitaxel (PTX) in pancreatic cancer treatment. Exosomes were successfully isolated from MSC-conditioned media using the tangential flow filtration (TFF) and ultracentrifuge, and the two approaches were quantitatively evaluated for isolation efficiency, yield, and scalability. The vesicles were characterized by the presence of tetraspanin surface markers (CD63 and CD81), confirming their exosomal nature, and by CD90, confirming their mesenchymal origin. PTX-loaded exosomes with high drug entrapment efficiency were prepared using a sonication method. PTX was selected as the model chemotherapeutic agent due to its proven efficacy against pancreatic and other solid tumors. The drug-loaded exosomes were further characterized for particle size, size distribution, zeta potential, morphology, drug loading

efficiency, and in vitro drug release profile. An extended safety assessment was conducted to address potential risks associated with MSC-derived vesicles, including a long-term analysis of epithelial–mesenchymal transition (EMT) marker expression to determine whether prolonged exposure induced phenotypic changes in cancer cells. Finally, their cytotoxic potential was evaluated in both 2D and 3D pancreatic cancer cell models to assess the therapeutic promise of MSC-derived exosomes as an effective nanocarrier for pancreatic cancer therapy. Accordingly, this study was designed as a preclinical in vitro proof-of-concept investigation, with a primary focus on manufacturing feasibility, safety, and tumor-relevant efficacy as a foundation for subsequent in vivo studies.

2. Materials and Methods

2.1. Materials

PTX (purity 99.96%) was procured from MedChemExpress (Monmouth Junction, NJ, USA). Mesenchymal stem cell medium (MSCM) was purchased from ScienCell Research Laboratories (Carlsbad, CA, USA). Penicillin–streptomycin solution, fetal bovine serum (FBS), and trypsin–EDTA (0.25%) were obtained from Sigma-Aldrich (St. Louis, MO, USA). HPLC-grade methanol, acetonitrile, and water were obtained from BDH Chemicals (Dawsonville, GA, USA). Dulbecco's phosphate-buffered saline (DPBS) and Dulbecco's Modified Eagle Medium (DMEM) were purchased from Corning (Manassas, VA, USA). A tangential flow filtration (TFF) unit was procured from Novus Biologicals (Centennial, CO, USA). APC anti-human CD81, FITC anti-human CD90, PE anti-human CD63 antibodies, and N-cadherin–PE were purchased from Thermo Fisher Scientific (Carlsbad, CA, USA). E-cadherin–PE was purchased from Cell Signaling Technology (Danvers, MA, USA).

2.2. Cell Culture

Bone marrow-derived human MSCs were purchased from ScienCell Research Laboratories (Carlsbad, CA, USA) and cultured in MSCM supplemented with 1% MSC growth supplement, 5% FBS, and 1% penicillin–streptomycin. MSCs at passages 2–7 were used for all experiments. Human pancreatic cancer cell lines PANC-1 and BxPC-3, along with the non-tumorigenic pancreatic epithelial cell line HPNE, were obtained from the American Type Culture Collection (ATCC, Manassas, VA, USA). PANC-1 and BxPC-3 cells were cultured in DMEM with 10% FBS and 1% penicillin–streptomycin. HPNE cells were grown in a specialized medium containing 5% FBS, 75% glucose-free DMEM, 25% Medium M3 Base, 10 ng/mL human recombinant EGF, 5.5 mmol/L D-glucose (1 g/L), and 750 ng/mL puromycin. All cell cultures were kept at 37 °C in a humidified incubator with 5% CO₂.

2.3. Isolation of Exosomes

Initially, exosomes were isolated from MSC-conditioned media using differential ultracentrifugation and TFF. For both methods, 4×10^6 MSCs were seeded into 182 cm² flasks and cultured for 24 h in complete MSCM. The medium was then replaced with serum-free MSCM, and conditioned media were collected after 24 h. For ultracentrifugation, conditioned media were centrifuged at 4000 rpm for 15 min to remove cellular debris, followed by filtration through a 0.22 µm membrane [³⁷]. The resulting supernatant was transferred to ultracentrifuge tubes and centrifuged at 100,000× *g* for 90 min (Beckman Coulter, Indianapolis, IN, USA). The translucent exosome pellet obtained was resuspended in 1 mL of DPBS and stored at –80 °C until further use.

The TFF system was operated according to the manufacturer's instructions. Briefly, conditioned media were prefiltered through a 0.45 µm mixed cellulose ester (MCE) membrane to remove cellular debris and other large particles. The filtrate was then processed through the TFF unit equipped with a 5 nm pore-size filter, enabling selective retention and concentration of EVs. Once the retentate volume was reduced to approximately 4 mL, the exosome-enriched fraction was collected, transferred to centrifuge tubes, and stored at –80 °C until further analysis.

2.4. Protein Quantification

The total protein content of the isolated exosomes was quantified using a bicinchoninic acid (BCA) assay kit (Thermo Scientific, Waltham, MA, USA). Before analysis, exosomes were lysed in radioimmunoprecipitation assay (RIPA) buffer for 15 min on ice to ensure complete protein solubilization. The assay was performed in a 96-well microplate according to the manufacturer's protocol. Briefly, BCA reagents A and B were mixed at a 50:1 ratio to prepare the working reagent. A standard curve was generated using serial dilutions of bovine serum albumin (BSA) as a reference protein. Equal volumes of the working reagent were added to each well containing standards or samples, and the mixture was incubated at 37 °C for 30 min. Absorbance was measured at 562 nm using a microplate reader, and total protein concentrations were determined from the BSA standard curve. All measurements were performed in triplicate to ensure reproducibility.

2.5. Exosome Marker Detection

The identity of exosomes was validated by assessing the expression of the tetraspanins surface markers CD63 and CD81. Additionally, surface expression of the MSC-specific marker CD90 was used to confirm their mesenchymal origin. Briefly, exosomes equivalent to 1 mg of total protein were incubated with PE anti-human CD63, APC anti-human CD81, or FITC anti-human CD90 antibodies for 30 min in the dark at room temperature. Following incubation, flow buffer was added to bring the total volume to 500 µL. Samples were analyzed immediately on a Cytoflex S (Beckman) flow cytometer in the NDSU Dr. Thomas Glass Biotech Innovation Core. Unstained exosomes served as negative controls. For each sample, approximately 10,000 events were recorded.

2.6. Preparation of PTX-Loaded Exosomes

PTX was encapsulated into exosomes using an optimized ultrasonication-assisted loading method. A PTX stock solution (100 mg/mL) was prepared in dimethyl sulfoxide (DMSO). Exosomes corresponding to 1 mg of total protein were resuspended in 2 mL of DPBS containing 90 µL of DMSO. Subsequently, 10 µL of the PTX stock solution (equivalent to 1 mg PTX) was added to the exosome suspension. The mixture was subjected to ultrasonication for five cycles of 20 s ON and 2 min OFF at 50 Hz, followed by incubation at 37 °C for 1 h to promote drug incorporation and membrane recovery. A large molar excess of drug relative to exosome protein was used to ensure that the hydrophobic drug had ample opportunity to partition into the exosomal membrane during ultrasonication.

Unencapsulated PTX and insoluble debris generated during ultrasonication were removed by centrifuging the formulation at 3200× *g* for 10 min. This process selectively pelleted precipitated free drug and particulate impurities, leaving PTX-loaded exosomes in the supernatant. The supernatant was subsequently centrifuged at 21,000× *g* for 30 min to pellet the PTX-loaded exosomes, while soluble, non-encapsulated PTX remained in the supernatant and was discarded. The exosome pellet was resuspended in DPBS and stored at 4 °C until further analysis.

2.7. Particle Size, Charge, and Morphology

The particle size, polydispersity index (PDI), and zeta potential of exosomes were determined by Malvern Zetasizer Nano ZS instrument (Malvern Instruments, Malvern, UK). Exosomes were dispersed in deionized water at a concentration of 0.1 mg/mL. Size and PDI were measured by dynamic light scattering (DLS) at a fixed detection angle of 90°, while zeta potential was determined using a Malvern Panalytical folded capillary cell. Particle size distribution and concentration were further analyzed by nanoparticle tracking analysis (NTA) using a NanoSight NS300 (Malvern Instruments, Malvern, UK). Samples were diluted in DPBS to achieve 1×10^6 – 1×10^9 particles/mL and analyzed in triplicate, with data processed using NanoSight NTA software (version 3.4).

Morphology was examined by transmission electron microscopy (TEM). A small aliquot of exosome suspension was deposited onto a carbon-coated copper grid, adsorbed for 2 min, and negatively stained with 0.1% (w/v) phosphotungstic

acid (pH 7.0–8.0). After air-drying, images were acquired using a JEOL JEM-2100 high-resolution TEM (JEOL USA, Peabody, MA, USA) at the North Dakota State University Electron Microscopy Core Facility (Fargo, ND, USA).

2.8. Drug Loading and Encapsulation Efficiency

PTX loading efficiency was evaluated by quantifying both PTX and exosomal protein concentrations using HPLC and the BCA protein assay, respectively. In brief, 50 μ L of the PTX-loaded exosome formulation was mixed with 950 μ L of acetonitrile to facilitate PTX extraction and protein precipitation. The mixture was incubated at room temperature for 30 min, followed by centrifugation at 17,000 $\times g$ for 10 min. The resulting supernatant was collected for PTX analysis, while the protein pellet was resuspended in DPBS and analyzed for protein content using the BCA method.

For HPLC analysis, samples (10 μ L injection volume) were run on a reverse-phase Accucore C18 column (100 mm $\times$ 2.1 mm, 2.6 μ m particle size). The mobile phase consisted of 10 mM ammonium acetate buffer (pH 4) and acetonitrile in a 1:1 (v/v) ratio, delivered isocratically at a flow rate of 0.3 mL/min [38]. Detection was performed at 254 nm using a Diode Array Detector (DAD) detector (Agilent Technologies, Santa Clara, CA, USA). The total run time was 10 min, with PTX eluting at approximately 2.45 min.

The drug loading and encapsulation efficiency were determined using the following equations.

2.9. In Vitro PTX Release from Exosomes

The release profile of PTX from exosomes was evaluated using a dialysis-based diffusion method. An exosomal suspension containing PTX equivalent to 1 mg was dispersed in deionized water and loaded into a Spectra-Por® Float-A-Lyzer® G2 dialysis membrane (molecular weight cut-off: 3.5–5 kDa; Spectrum Laboratories, Rancho Dominguez, CA, USA). The dialysis bag was then immersed in 30 mL of release medium composed of DPBS supplemented with 1% (v/v) Tween 80 to maintain sink conditions. The system was maintained at 37 °C under constant agitation at 100 rpm using an orbital shaker (MAXQ 4000, Thermo Scientific, Marietta, OH, USA) as previously described [39,40]. At predetermined time points (1, 2, 6, 12, 24, 72, 120, 168, 240, and 336 h), 1 mL aliquots of the release medium were withdrawn and replaced with an equal volume of fresh pre-warmed medium. The concentration of PTX released into the medium was quantified by HPLC using our optimized method [38]. The PTX release data were also analyzed using KinetDS 2.0 software to determine the drug release kinetics.

2.10. In Vitro Biocompatibility of Exosomes

The biocompatibility of blank (drug-free) exosomes was evaluated by assessing their effects on cell proliferation in human pancreatic cancer cell lines (PANC-1 and BxPC-3) and non-tumorigenic pancreatic ductal epithelial cells (HPNE). Cells were seeded into 96-well plates at a density of 5000 cells per well and allowed to adhere for 24 h in complete growth medium. Afterward, blank exosomes were added at final concentrations ranging from 10⁴ to 10¹⁰ particles/mL. Following 72 h of treatment, cell proliferation was quantified using the MTS assay by aspirating the culture medium and adding 200 μ L of freshly prepared MTS solution (growth media:MTS:PMS, 100:20:1) to each well. Plates were incubated at 37 °C for 1 h, and the absorbance was measured at 490 nm using a SpectraMax M5 microplate reader (Molecular Devices, Sunnyvale, CA, USA). Cells cultured in exosome-free growth media served as controls for relative cell viability calculations. For all samples, background absorbance was corrected by subtracting the mean reading from wells containing only the MTS reagent.

2.11. Epithelial to Mesenchymal Transition Study

The effect of mesenchymal stem cell-derived exosomes on EMT in PDAC cells was assessed by measuring surface expression of the EMT markers E-cadherin and N-cadherin using flow cytometry, as decreased E-cadherin and increased N-cadherin indicate a shift toward a mesenchymal phenotype. BxPC-3 and PANC-1 cells were treated with drug-free

exosomes for 7 or 14 days to evaluate whether prolonged exposure induced phenotypic changes. For the 7-day treatment, 150,000 cells were seeded per well in 6-well plates, whereas 10,000 cells per well were seeded for the 14-day condition to support extended culture. After each treatment period, cells were harvested, washed, and prepared for staining, with approximately 1×10^5 cells used per sample. Cells were incubated with fluorophore-conjugated antibodies (E-cadherin-PE and N-cadherin-PE) on ice and in the dark for 20 min, followed by washing to remove unbound antibody. Stained cells were resuspended in flow buffer and immediately analyzed to quantify exosome-induced changes in epithelial versus mesenchymal marker expression.

2.12. Cellular Uptake of Exosomes

To evaluate the time-dependent cellular uptake of exosomes, PANC-1 cells were seeded in 24-well plates and allowed to adhere overnight. The cells were then treated with DiI-labeled exosomes and incubated for predetermined time points (0.5, 1, 2, 4, and 6 h). Following incubation, cells were washed twice with DPBS to remove unbound exosomes, trypsinized, and lysed using RIPA buffer. The lysates were lyophilized for 48 h, then extracted to elute DiI with methanol, which was quantified using a SpectraMax M5 microplate reader (Molecular Devices, Sunnyvale, CA, USA) with excitation and emission wavelengths of 549 nm and 567 nm, respectively.

To further visualize cellular uptake, exosomes were fluorescently labeled with the lipophilic dye DiI according to the manufacturer's protocol. Cells were incubated with DiI-labeled exosomes (equivalent to 1 mg/mL exosomal protein) for the same time intervals and then washed gently with DPBS to remove excess dye. Cells were fixed with 4% paraformaldehyde for 15 min and counterstained with DAPI to visualize nuclei. Fluorescent images were captured using a Leica DMI8 fluorescence microscope (Leica Microsystems, Wetzlar, Germany), enabling qualitative assessment of exosome internalization and intracellular distribution.

2.13. In Vitro Cytotoxicity

The in vitro cytotoxicity of PTX-loaded exosomes was assessed using a cell proliferation assay and compared with PTX solution. PANC-1 and BxPC-3 were seeded into 96-well plates at a density of 5000 cells per well and allowed to adhere for 24 h. Cells were then treated with increasing concentrations of PTX (0.01–1000 nM) or with PTX-loaded exosomes containing equivalent drug doses. After 72 h of exposure, cell viability was evaluated using the MTS assay as previously described. Half-maximal inhibitory concentration (IC_{50}) values were determined by nonlinear regression analysis using GraphPad Prism (version 10.6.0) (GraphPad Software, San Diego, CA, USA), and dose–response curves were generated accordingly.

2.14. Colony Formation Assay

PANC-1 cells were trypsinized, counted, and seeded at a density of 1000 viable cells per well in 6-well plates and allowed to attach overnight. The following day, the medium was replaced with fresh growth containing 5 nM PTX or PTX-loaded exosomes containing an equivalent PTX dose. In contrast, cells treated with drug-free growth media served as a vehicle control group. After treatment, cells were maintained for 14 days under standard culture conditions to allow colony formation. Colonies were then fixed with 4% paraformaldehyde for 15 min at room temperature, washed with DPBS, stained with 0.5% (w/v) crystal violet in 10% ethanol for 30 min, rinsed to remove excess stain, and air-dried before being imaged and quantified with ImageJ (version 1.54k).

2.15. In Vitro Cytotoxicity Against Spheroid Model

PANC-1 spheroids were generated to evaluate the antitumor activity of PTX-loaded exosomes under 3D culture conditions. Briefly, 2500 cells per well were added to ultra-low attachment, round-bottom 96-well plates, and the plates were centrifuged at $250 \times g$ for 10 min to promote uniform cell aggregation at the center of each well. Spheroids were maintained under standard culture conditions until their average diameter exceeded 500 μm , at which point treatment

was initiated and designated as Day 0. Treatment groups received PTX-loaded exosomes or a free PTX solution at final concentrations of 50 and 100 nM. Following treatment, spheroid growth was monitored for 3 days using an AmScope IN300T-FL inverted fluorescence microscope (Irvine, CA, USA). Spheroid diameters were measured daily, and changes in calculated spheroid volume were used as a surrogate indicator of tumor growth and therapeutic response.

2.16. Statistical Analysis

Data are expressed as mean $\pm$ SD. Statistical significance was evaluated using two-tailed Student's *t*-tests or one-way ANOVA, and differences were considered significant at $p < 0.05$.

3. Results

3.1. Isolation and Characterization of Exosomes

Exosomes were isolated from MSC-conditioned media using two approaches, differential ultracentrifugation and TFF. Both methods were initially assessed to compare their yields and efficiencies. Data from nanoparticle tracking analysis (NTA), supported by BCA protein quantification, demonstrated that TFF consistently produced substantially higher exosome yields, approximately 8–9-fold greater than those obtained by ultracentrifugation. Specifically, the total particle yield obtained using TFF was 5.36×10^{11} particles, whereas ultracentrifugation yielded 6.02×10^{10} particles. Based on a total of approximately 8×10^6 MSCs harvested after conditioned media collection, TFF produced an estimated 67,000 exosomes/MSC, compared with only ~7500 exosomes/MSC for ultracentrifugation. A similar pattern was observed for total protein recovery. TFF yielded approximately 52.3 mg of exosomal protein, while UC recovered only ~5.9 mg from the same number of cells. Given the markedly higher yield, improved reproducibility, and faster processing time associated with TFF, all subsequent experiments used exosomes isolated exclusively by TFF.

3.2. Exosome Marker Detection

The identity and source of the isolated vesicles were established using flow cytometry. Staining for CD63, CD81, and CD90 yielded significantly higher fluorescence signals in exosome samples than in unstained controls (^[Figure 1]). The presence of CD63 and CD81 markers confirmed that the vesicles were exosomes, while positive staining for CD90 indicated that they were derived from MSCs.

Flow cytometric characterization of MSC-derived exosomes. Exosomes were characterized for surface expression of exosomal tetraspanins CD63, CD81, and the MSC-associated marker CD90.

3.3. Preparation and Characterization of PTX-Loaded Exosomes

The physicochemical characteristics of blank and PTX-loaded exosomes were assessed using NTA, DLS, zeta potential analysis, and TEM. NTA revealed a predominant particle population within the expected range, with blank exosomes showing a major peak at approximately 117 nm for ultracentrifugation and 162 nm for TFF (^[Figure S1]). DLS measurements confirmed these findings, showing mean diameters of 160 ± 16.4 nm for blank exosomes and 168.6 ± 2.4 nm for PTX-loaded exosomes, indicating a slight increase in size following drug incorporation (^[Figure 2]A,B and ^[Table 1]). Both formulations exhibited stable negative surface charges, with zeta potentials of -28.4 ± 2.4 mV and -26.5 ± 3.3 mV, respectively (^[Figure 2]C,D and ^[Table 1]). TEM imaging further demonstrated the characteristic spherical morphology of exosomes and particle sizes consistent with those determined by NTA and DLS (^[Figure 2]E,F). PTX loading was efficient, yielding an encapsulation efficiency of $31.3 \pm 2.0\%$ and a loading capacity of $23.9 \pm 1.1\%$, confirming successful drug incorporation (^[Table 1]).

Physicochemical characterization of exosomes. (A) Particle size distribution of blank exosomes as measured by dynamic light scattering. (B) Particle size distribution of PTX-loaded exosomes. (C) Zeta potential distribution of blank

exosomes. (D) Zeta potential distribution of PTX-loaded exosomes. (E) Transmission electron microscopy (TEM) image of (E) blank exosomes and (F) PTX-loaded exosomes. Scale bar = 100 nm.

Physicochemical characterization of drug-free (Blank Exo) and PTX-loaded exosomes (Exo-PTX). Data represent mean $\pm$ SD ($n = 6$).

3.4. In Vitro Drug Release Study

The release kinetics of PTX from PTX-loaded exosomes were evaluated using a dialysis-based method under physiological sink conditions. As shown in [Figure 3], PTX exhibited a biphasic release profile. An initial burst phase occurred within the first 24 h, during which a substantial portion of the encapsulated drug diffused out of the exosomal matrix. This was followed by a sustained, gradual release over the 7-day study period, indicating continued diffusion of PTX through the exosomal membrane.

In vitro drug release profile of PTX-loaded exosomes in 1% (v/v) Tween 80 supplemented DPBS at 37 °C. Data represent mean $\pm$ SD ($n = 4$).

To elucidate the kinetics of PTX release, the data were analyzed using KinetDS 2.0. The kinetic model with the highest correlation coefficient (r^2) was selected to provide the most accurate representation of the release mechanism [41]. Among the models evaluated, the Weibull model demonstrated the strongest correlation with the experimental data ($R^2 = 0.9432$) and accurately described both the initial burst release and the subsequent sustained-release phase ([Table 2]). In contrast, the inclusion of a lag time resulted in a lower fit ($R^2 = 0.8151$), indicating that PTX release commenced immediately upon exposure to the release medium. These results highlight the potential of exosomes as controlled-release nanocarriers for hydrophobic drugs such as PTX, emphasizing their applicability in advanced drug delivery systems.

Mathematical models for in vitro release of PTX from MSC-derived exosomes at pH 7.4 and their corresponding R^2 values.

3.5. In Vitro Biocompatibility of Exosomes

The biocompatibility of blank exosomes was assessed by examining their effects on the proliferation of multiple cell lines across a wide range of concentrations (10^4 – 10^{10} exosomes/mL). As shown in [Figure 4], exposure to blank exosomes did not significantly alter cell viability in any of the tested cell lines, including HPNE. Cell proliferation remained comparable to that of untreated controls at all concentrations, with no indication of cytotoxicity or proliferation stimulation. These findings demonstrate that blank exosomes are well tolerated and exhibit excellent biocompatibility, even at the highest doses evaluated.

In vitro biocompatibility of drug-free exosomes using pancreatic cancer cells PANC-1 and BxPC-3 and non-tumorigenic pancreatic epithelial cell line HPNE. Data represent mean $\pm$ SD ($n = 6$).

3.6. Epithelial to Mesenchymal Transition Study

Flow cytometry was performed to assess whether MSC-derived exosomes influence the epithelial–mesenchymal transition of PANC-1 and BxPC-3 cells. Following treatment with MSC-derived exosomes for 7 or 14 days, neither cell line exhibited notable changes in cadherin expression ([Figure 5]). The fluorescence intensity profiles of N-cadherin and E-cadherin in treated groups closely overlapped with those of their respective controls, with no detectable shifts in peak position or distribution. These findings indicate that MSC-derived exosomes did not alter the epithelial or mesenchymal characteristics of PANC-1 or BxPC-3 cells under the tested conditions.

Effect of exosomal treatment on epithelial to mesenchymal transition (EMT) of pancreatic cancer cells. Histogram representation of E-cadherin and N-cadherin expression in PANC-1 (A,B) and BxPC-3 (C,D) cells after treatment with MSC-derived exosomes for 7 days or 14 days. Unstained cells (red) are included as negative controls to confirm gating and baseline fluorescence.

3.7. Cellular Uptake of PTX-Loaded Exosomes

The internalization of DiI-labeled exosomes by PANC-1 cells was quantified over 6 h using a Spectra-Max M5 microplate reader. As shown in [Figure 6], cellular uptake increased steadily during the first 4 h of incubation, indicating rapid and efficient internalization of exosomes. The percentage of intracellular DiI reached $30.8 \pm 4.9\%$ at 0.5 h, $40.7 \pm 8.3\%$ at 1 h, $52.8 \pm 5.2\%$ at 2 h, and peaked at $61.0 \pm 7.0\%$ at 4 h. Beyond this time point, there was no further increase in cell uptake, suggesting that saturation was achieved. Overall, these data indicate that PANC-1 cells rapidly internalize DiI-labeled exosomes, with maximal uptake occurring within the first few hours of exposure.

Cellular uptake of DiI-labeled exosomes in PANC-1 cells. Data represent mean $\pm$ SD ($n = 4$).

Time-dependent cellular uptake of DiI-labeled MSC-derived exosomes is depicted in [Figure 7]. The images across 4 progressive time points show the gradual uptake and internalization of exosomes by PANC-1 cells.

Representative fluorescence microscopy images of DiI-labeled exosomes (red) by PANC-1 cells over time (1, 2, 4, and 6 h). Nuclei were stained with DAPI (blue), and corresponding brightfield images are shown for morphological context. Scale bar = 100 μ m.

3.8. In Vitro Cytotoxicity of PTX-Loaded Exosomes

The cytotoxic effects of PTX-loaded exosomes were evaluated in PANC-1 and BxPC-3 pancreatic cancer cell lines and compared with those of free PTX solution. As shown in [Figure 8], both formulations produced dose-dependent reductions in cell viability; however, PTX-loaded exosomes consistently demonstrated greater potency across the tested concentration range. In PANC-1 cells, the IC_{50} of the exosomal formulation (19.29 nM) was lower than that of free PTX (22.44 nM) ([Figure 8]A). A similar trend was observed in BxPC-3 cells, where PTX-loaded exosomes exhibited a markedly lower IC_{50} value (7.55 nM) than free PTX (12.48 nM), indicating enhanced cytotoxic activity ([Figure 8]B). The leftward shift in the dose–response curves further demonstrate improved efficacy of exosome-mediated PTX delivery. Collectively, these findings indicate that encapsulation within exosomes enhances the antiproliferative effects of PTX in pancreatic cancer cells.

In vitro cytotoxicity of PTX-loaded exosomes (Exo-PTX) compared to PTX solution. Dose–response curves for cytotoxicity in (A) PANC-1 and (B) BxPC-3 cells ($n = 6$).

3.9. Colony Formation Assay

The clonogenic potential of PANC-1 cells following PTX treatment was evaluated using a colony formation assay. As shown in [Figure 9]A, untreated control cells produced abundant colonies, which were normalized to 100%. Treatment with free PTX solution (PTX Sol, 5 nM) resulted in a significant reduction in colony formation to 38.9% of control levels ($p < 0.005$) ([Figure 9]B). A further and more pronounced inhibitory effect was observed with PTX-loaded exosomes (PTX-Exo, 5 nM equivalent), which reduced colony numbers to 17.7% of control ($p < 0.005$ vs. CTRL and PTX Sol). These findings demonstrate that exosomal delivery substantially enhances the anticlonogenic activity of PTX, thereby improving therapeutic efficacy compared with the conventional formulation.

Effect of Exo-PTX on colony formation potential of PANC-1 cells. (A) Representative images of colonies formed in control, PTX Sol (5 nM), and Exo-PTX (5 nM equivalent) treated cells after 14 days. (B) Quantitative evaluation of

colonies following various treatments. Data represent mean $\pm$ SD ($n = 6$). *** $p < 0.005$.

3.10. In Vitro Cytotoxicity Against Spheroid Model

The anticancer efficacy of PTX-loaded exosomes was further examined using a 3D PANC-1 spheroid model. Both Exo-PTX and free PTX displayed dose-dependent inhibition of spheroid growth; however, Exo-PTX produced a more rapid therapeutic response during the early stages of treatment. As shown in [Figure 10] and [Figure S2], control spheroids continued to grow throughout the study, reaching approximately 126% of their initial volume by Day 3. In contrast, spheroids treated with PTX formulations began to shrink from Day 1.

*In vitro cytotoxicity profile of PTX solution and Exo-PTX using PANC-1 spheroid. (A) Brightfield images of spheroids taken on days 0, 1, 2, and 3. Scale bar = 500 μ m. (B) Growth chart of spheroids over 3 days following treatment initiation. Data represent mean $\pm$ SD ($n = 5$). * $p < 0.05$ from control and # $p < 0.05$ from PTX solution. The control group for both [Figure 10] and [Figure S2] is the same.*

Across the tested concentrations (50 nM and 100 nM), the Exo-PTX group exhibited an earlier and more pronounced reduction in spheroid size compared to free PTX. On Day 2, spheroids treated with Exo-PTX (50 nM) had decreased to $80.6 \pm 3.3\%$ of their initial volume, whereas those treated with free PTX (50 nM) decreased only to $85.6 \pm 6.1\%$, reflecting a slower onset of cytotoxic activity ([Figure 10]B). By Day 3, Exo-PTX–treated spheroids increased slightly to $85.6 \pm 4.6\%$, while free PTX–treated spheroids increased a lot more, $98.8 \pm 4.6\%$ of its original size, demonstrating significantly enhanced efficacy of Exo-PTX ($n = 5$, $p < 0.05$ vs. free PTX on Day 3). At 100 nM, a similar trend was observed ([Figure S2]), and on Day 2 spheroids treated with Exo-PTX were reduced to $72.6 \pm 7.0\%$ of their initial volume compared with $76.1 \pm 4.4\%$ for free PTX ([Figure S2B]). By Day 3, Exo-PTX–treated spheroids remained substantially smaller at $69.3 \pm 6.3\%$, whereas those treated with free PTX showed less sustained shrinkage at $78.8 \pm 6.8\%$, further confirming the superior and sustained antitumor efficacy of Exo-PTX.

4. Discussion

This study demonstrates the feasibility and therapeutic potential of mesenchymal stem cell–derived exosomes as nanocarriers for PTX delivery in pancreatic cancer. By employing TFF for exosome isolation, we achieved significantly higher particle and protein yields than with differential ultracentrifugation, consistent with prior reports that highlight TFF as a scalable, gentle method that preserves vesicle integrity while providing superior recovery [42]. Furthermore, the near-identical protein-to-particle ratios observed for both methods suggest broadly comparable vesicle-associated composition. While we recognize that yield-based analyses alone cannot completely exclude some degree of co-isolated non-vesicular materials, including protein aggregates and lipoproteins, the consistency of these ratios suggests that the increased recovery with TFF reflects a proportional enrichment of vesicle-associated components. In this context, the results support TFF as an effective and scalable approach for extracellular vesicle isolation. The high yield obtained here is particularly important for translational applications, where sufficient exosome production remains a key bottleneck to clinical implementation [43]. Additional purity-focused analyses in future studies will further refine and extend these observations.

The exosomes isolated in this study exhibited hallmark structural and molecular characteristics, including size distributions within the expected nanoscale range, negative zeta potentials, and expression of CD63, CD81, and MSC-specific CD90. These results demonstrate that MSC-derived exosomes were effectively isolated. PTX loading via ultrasonication achieved favorable encapsulation efficiency without compromising vesicle morphology, demonstrating the utility of this approach for incorporating hydrophobic chemotherapeutics. The slight increase in size and the modest change in surface charge following drug loading further supported the successful incorporation of PTX into the exosome.

The biphasic release profile of PTX from exosomes, characterized by an initial burst followed by a sustained release over seven days, suggests a therapeutically advantageous delivery system that balances early drug availability with prolonged exposure. Controlled release is particularly desirable in pancreatic cancer, where poor vascularization, dense stromal barriers, and high interstitial pressure reduce effective intracellular drug accumulation. The extended release observed in our study supports the potential for enhanced drug retention within the tumor microenvironment.

The Weibull model emerged as the most appropriate framework for describing PTX release from MSC-derived exosomes, as it accounts for the complex, nonlinear release behavior characteristic of lipid bilayer-based nanocarriers. Unlike classical kinetic models that assume a single dominant mechanism, the Weibull function is empirical and flexible, making it well-suited for systems where a combination of diffusion, partitioning, and carrier–drug interactions governs drug release [44].

MSC-derived exosomes also exhibited excellent biocompatibility, as blank exosomes did not induce cytotoxicity or proliferative changes across non-tumorigenic or cancerous pancreatic cell lines. Moreover, long-term exposure to blank exosomes did not alter cadherin expression in PANC-1 or BxPC-3 cells, indicating no induction of epithelial–mesenchymal transition. This is an important safety consideration, given ongoing concerns that MSC-derived products may modulate tumor progression in some contexts [45, 46].

Encapsulation of PTX enhanced anticancer efficacy across multiple experimental models. PTX-loaded exosomes reduced IC₅₀ values in both PANC-1 and BxPC-3 cells, demonstrating improved potency over the free drug. Improved clonogenic inhibition and more rapid reduction in spheroid volume further highlight the benefits of exosomal PTX delivery for targeting aggressive, drug-resistant tumor populations. Enhanced cellular uptake of PTX-loaded exosomes, confirmed by quantitative HPLC analysis, likely underlies the superior cytotoxic effects observed. The rapid and efficient uptake reaching saturation by 4 h supports the concept that exosomes leverage endogenous endocytic pathways to achieve more effective intracellular delivery compared to free PTX.

The results observed in the 3D spheroid model are particularly encouraging, as spheroids recapitulate key structural and biochemical features of tumor architecture, including hypoxia, nutrient gradients, and extracellular matrix deposition. The ability of exosomal PTX to induce earlier and more pronounced spheroid shrinkage than free PTX suggests improved penetration, consistent with prior studies showing that biological vesicles traverse dense tumor matrices more effectively than synthetic nanocarriers [47, 48].

Overall, the present research provides a systematically integrated evaluation of scalable exosome manufacturing, extended safety profiling, and tumor-relevant in vitro efficacy within a single, unified framework. This study quantitatively compares TFF and ultracentrifugation in terms of exosome yield and scalability, demonstrates that prolonged exposure to MSC-derived exosomes does not alter EMT, and directly compares the anticancer efficacy of free PTX and PTX-loaded exosomes in both 2D monolayer cultures and physiologically relevant 3D spheroid models. These advances address key translational barriers related to manufacturing feasibility, safety, and tumor-relevant efficacy, which are often considered separately in prior studies. However, comprehensive in vivo assessments in orthotopic pancreatic tumor models remain essential to validate biodistribution, tumor homing, clearance kinetics, and therapeutic efficacy, as well as to further optimize exosome–tumor interactions through mechanistic and engineering-based approaches.

5. Conclusions

MSC-derived exosomes show strong potential as biocompatible and effective nanocarriers for delivering PTX in pancreatic cancer in vitro models. Exosomes isolated via tangential flow filtration showed high yield, structural integrity, and MSC-specific marker expression, while ultrasonication enabled efficient drug loading without compromising vesicle

quality. PTX-loaded exosomes provided sustained drug release, enhanced cellular uptake, and had superior anticancer efficacy in both 2D and 3D models compared to free PTX. Blank exosomes were non-toxic and did not alter cellular phenotype, confirming their safety. Overall, these results highlight MSC-derived exosomes as a promising platform for improving PTX delivery in pancreatic cancer, while underscoring the need for comprehensive in vivo studies to fully establish their translational potential.

Acknowledgments

During the preparation of this manuscript, the author(s) used ChatGPT 5.2 for the purposes of text editing, such as grammar, structure, spelling, and punctuation. The authors have reviewed and edited the output and take full responsibility for the content of this publication.

Abbreviations

The following abbreviations are used in this manuscript:

Supplementary Materials

The following supporting information can be downloaded at: [<https://www.mdpi.com/article/10.3390/biom16020269/s1>], Figure S1: Nanoparticle tracking analysis (NTA) showing the size range of MSC-derived exosomes; Figure S2: In vitro cytotoxicity profile of PTX solution (100 nM) and Exo-PTX (100 nM PTX equivalent) using PANC-1 spheroid.

Author Contributions

A.B.: Methodology, Validation, Formal analysis, Investigation, Data Curation, Writing—Original Draft, Writing—Review & Editing, Visualization. A.G.: Methodology, Investigation, Writing—Review & Editing. P.M.G.: Methodology, Investigation, Writing—Review & Editing. S.T.: Methodology, Investigation, Writing—Review & Editing. Y.C.: Resources and Writing—Review & Editing. B.L.: Conceptualization, Resources, Writing—Original Draft, Writing—Review & Editing, Visualization, Supervision, Project administration, Funding acquisition. All authors have read and agreed to the published version of the manuscript.

Institutional Review Board Statement

Not applicable.

Informed Consent Statement

Not applicable.

Data Availability Statement

The data presented in this study are contained within the article and [[Supplementary Materials](#)].

Conflicts of Interest

The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Funding Statement

This research was funded by National Institute of General Medical Sciences (NIGMS) COBRE award 2P20 GM109024 and Engebretson Family Research Fund.

Footnotes

References

Associated Data

Supplementary Materials

Data Availability Statement

The data presented in this study are contained within the article and ^[Supplementary Materials].